

Experiment: 2

Aim:

To study source code version control, centralized versus distributed version control, the Git workflow, Git installation, and important Git commands.

1 Version Control

Version control is a system used to track and manage changes made to source code over time. It maintains different versions of a project and allows developers to recover previous versions when required.

Advantages

- Tracks changes in source code.
- Maintains project history.
- Allows recovery of previous versions.
- Supports merging and branching.
- Helps multiple developers work together.

2 Types of Version Control Systems

There are two main types:

Centralized Version Control System (CVCS)

In CVCS, the source code and its history are stored on a central server. Developers access the central repository to work on the project.

Examples: SVN, CVS.

Distributed Version Control System (DVCS)

In DVCS, every developer has a complete copy of the repository and its history on their local system. Most operations can be performed locally.

Examples: Git, Mercurial.

CVCS vs. DVCS

CVCS	DVCS
Central repository	Complete local repository
Higher server dependency	Lower server dependency
Local operations are limited	Most operations can be performed locally
Example: SVN	Example: Git

3 Git

Git is a free and open-source Distributed Version Control System (DVCS) used to track changes in source code.

Git provides important features such as:

- Version tracking
- Branching
- Merging
- Local repositories
- Remote repositories
- Collaboration

Git and GitHub are different: Git is a version control system, whereas GitHub is a platform used for hosting and collaborating on Git repositories.

4 Git Working Areas

Git mainly works with three areas:

1. **Working Directory:** The project folder where developers create and modify files.
2. **Staging Area:** An intermediate area where changes are selected before they are committed.
3. **Local Repository:** Stores committed changes along with the project's version history.

Basic Flow

Working Directory → Staging Area → Local Repository → Remote Repository

5 Git Workflow

The Git workflow consists of modifying files, selecting changes for staging, committing changes to the local repository, and synchronizing changes with a remote repository.

It allows developers to maintain a proper history of project changes and collaborate efficiently.

6 Branching

A branch is an independent line of development used to work on a particular feature or task without directly affecting the main codebase.

Importance of Branching

- Supports parallel development.
- Keeps the main branch stable.
- Isolates new features.
- Reduces risk to existing code.

- Makes collaboration easier.

Keep the main branch stable and perform feature development in separate branches.

7 Branching Strategies

GitFlow

GitFlow is a structured branching model that uses branches such as:

- Main/Master
- Develop
- Feature
- Release
- Hotfix

GitHub Flow

GitHub Flow is a simpler branching strategy where developers create feature branches, develop changes, review them, and merge them into the main branch.

8 Git Installation

Git can be installed on Windows using the Git installer. During installation, the required components, installation location, default editor, PATH settings, and other configuration options are selected. After installation, Git can be verified and configured with the user's identity.

9 Important Git Commands

1. Create repository
`git init`

2. Check status
`git status`

3. Add files
`git add .`

4. Commit changes
`git commit -m "Initial commit"`

5. Create branch
`git branch feature`

6. Switch branch
`git checkout feature`

7. Make changes
Edit project files

8. Stage changes
`git add .`

9. Commit
`git commit -m "Added feature"`

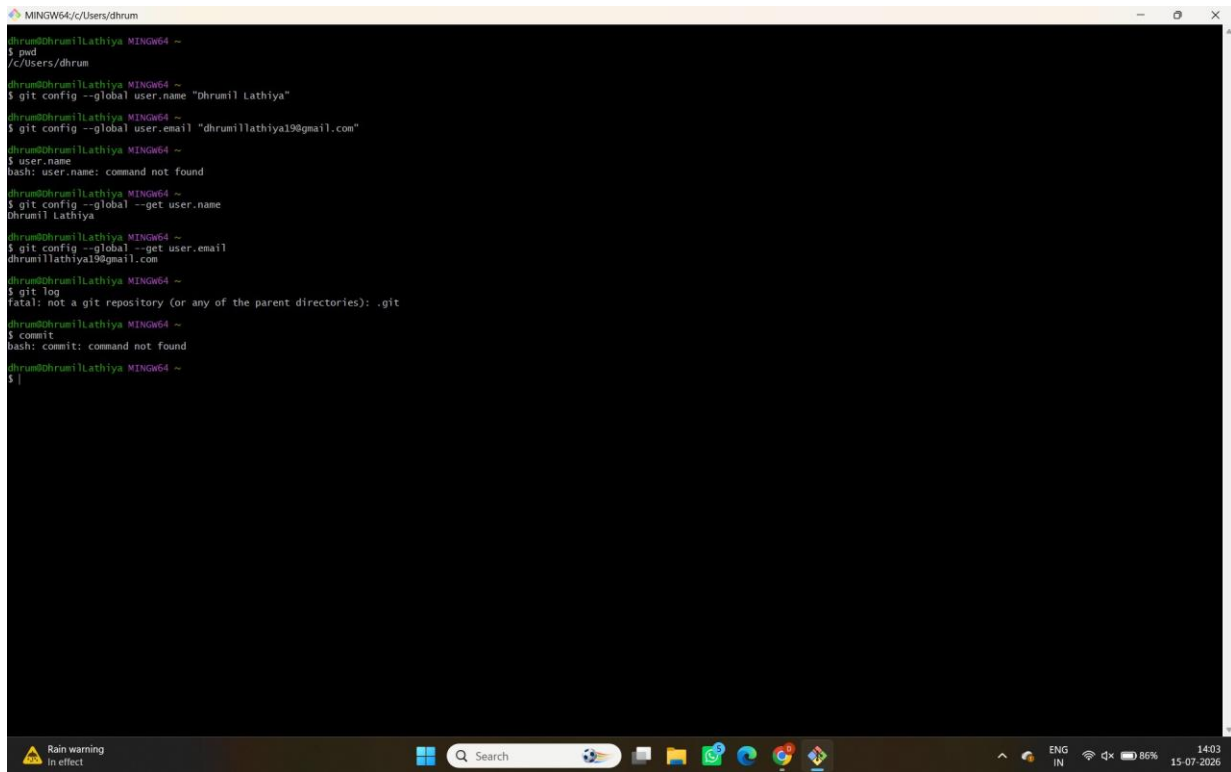
10. Switch to main
`git checkout main`

11. Merge feature
`git merge feature`

For a remote repository:

```
git remote add origin <repository-url>
git push origin main
git pull origin main
```

The following screenshots document Git identity configuration, directory navigation, repository initialization, commit history, and feature-branch development. They show staging and committing files, checking repository status, and errors encountered during navigation and branch switching.



```
MINGW64/c/Users/dhrum
dhrum@dhrumilathiya MINGW64 ~
$ pwd
/c/Users/dhrum
dhrum@dhrumilathiya MINGW64 ~
$ git config --global user.name "Dhrumil Lathiya"
dhrum@dhrumilathiya MINGW64 ~
$ git config --global user.email "dhrumilathiya19@gmail.com"
dhrum@dhrumilathiya MINGW64 ~
$ user.name
bash: user.name: command not found
dhrum@dhrumilathiya MINGW64 ~
$ git config --global --get user.name
Dhrumil Lathiya
dhrum@dhrumilathiya MINGW64 ~
$ git config --global --get user.email
dhrumilathiya19@gmail.com
dhrum@dhrumilathiya MINGW64 ~
$ git log
fatal: not a git repository (or any of the parent directories): .git
dhrum@dhrumilathiya MINGW64 ~
$ commit
bash: commit: command not found
dhrum@dhrumilathiya MINGW64 ~
$ !
```

Figure 1: Initial Git configuration and command errors.

Description: Sets and verifies the Git user identity; invalid commands and running git log outside a repository produce errors.

```
MINGW64/C:/Users/dhrum/OneDrive/Desktop
$ user.name
bash: user.name: command not found

dhrum@dhrumilLathiya MINGW64 ~
$ git config --global --get user.name
Dhrumil Lathiya

dhrum@dhrumilLathiya MINGW64 ~
$ git config --global --get user.email
dhrumilathiya19@gmail.com

dhrum@dhrumilLathiya MINGW64 ~
$ git log
fatal: not a git repository (or any of the parent directories): .git

dhrum@dhrumilLathiya MINGW64 ~
$ commit
bash: commit: command not found

dhrum@dhrumilLathiya MINGW64 ~
$ git commit
fatal: not a git repository (or any of the parent directories): .git

dhrum@dhrumilLathiya MINGW64 ~
$ cd Desktop/Car Dealership Inventory System_1
bash: cd: too many arguments

dhrum@dhrumilLathiya MINGW64 ~
$ cd Desktop
bash: cd: Desktop: No such file or directory

dhrum@dhrumilLathiya MINGW64 ~
$ ls
AndroidStudioProjects/  Documents/  'My Documents'0
Application Data0      Downloads/  NTUSER.DAT
Cisco Packet Tracer 8.2.1/  Favorites/  NTUSER.DAT{3c118866-b738-11ef-b2d7-f7f433b35654}.TM.bif
Contacts/              IdeaProjects/  Links/  NTUSER.DAT{3c118866-b738-11ef-b2d7-f7f433b35654}.TM.Container00000000000000000001.regtrans-ms
Cookies0               'Local Settings'0  NetHood/  NTUSER.DAT{3c118866-b738-11ef-b2d7-f7f433b35654}.TM.Container00000000000000000002.regtrans-ms
DOSBox-0.74/           Music/         OneDrive/  Postman/
                                PrintHood0
Recent0
'Saved Games'0
Searches/
SendTo0
SpringbootProjects/
'Start Menu'0
Templates0
Videos/
anaconda3/
ntuser.dat.LOG1
ntuser.dat.LOG2
ntuser.ini
wallet-public.pem
wallet.jks

dhrum@dhrumilLathiya MINGW64 ~
$ cd Desktop
bash: cd: Desktop: No such file or directory

dhrum@dhrumilLathiya MINGW64 ~
$ /C
$ cd OneDrive
dhrum@dhrumilLathiya MINGW64 ~/OneDrive
$ cd Desktop
dhrum@dhrumilLathiya MINGW64 ~/OneDrive/Desktop
$ cd Car Dealership Inventory System_1
bash: cd: too many arguments

dhrum@dhrumilLathiya MINGW64 ~/OneDrive/Desktop
$ cd CarDealershipInventorySystem_1
bash: cd: CarDealershipInventorySystem_1: No such file or directory

dhrum@dhrumilLathiya MINGW64 ~/OneDrive/Desktop
$ |
```

Figure 2: Locating the project directory in Git Bash.

Description: Navigates to Desktop through OneDrive; incorrect folder names and unquoted spaces cause directory-change errors.

```
dhrum@dhrumilLathiya MINGW64 ~
$ pwd
/c/Users/dhrum

dhrum@dhrumilLathiya MINGW64 ~
$ git config --global user.name "Dhrumil Lathiya"

dhrum@dhrumilLathiya MINGW64 ~
$ git config --global user.email "dhrumilathiya19@gmail.com"
```

Figure 3: Setting the global Git author identity.

Description: Uses `git config --global` to set the default author name and email for commits.

```

dhrum@DhrumilLathiya MINGW64 ~
$ git config --global --get user.name
Dhrumil Lathiya

dhrum@DhrumilLathiya MINGW64 ~
$ git config --global --get user.email
dhrumillathiya19@gmail.com

dhrum@DhrumilLathiya MINGW64 ~
$ git log
fatal: not a git repository (or any of the parent directories): .git

```

Figure 4: Verifying the identity and attempting to view commit history.

Description: Verifies the saved name and email; git log fails because the current folder is outside a Git repository.

```

dhrum@DhrumilLathiya MINGW64 ~
$ commit
bash: commit: command not found

dhrum@DhrumilLathiya MINGW64 ~
$ git commit
fatal: not a git repository (or any of the parent directories): .git

dhrum@DhrumilLathiya MINGW64 ~
$ cd Desktop\Car Dealership Inventory System_1
bash: cd: too many arguments

dhrum@DhrumilLathiya MINGW64 ~
$ cd Desktop
bash: cd: Desktop: No such file or directory

```

Figure 5: Commit-command and directory-path errors.

Description: Shows errors from using commit alone, running git commit outside a repository, and entering invalid paths.

```

dhrum@DhrumilLathiya MINGW64 ~
$ ls
AndroidStudioProjects/  Documents/  'My Documents'@
AppData/               Downloads/  NTUSER.DAT
'Application Data'@    Favorites/  NTUSER.DAT{3c118866-b738-11ef-b2d7
Cisco Packet Tracer 8.2.1'/  IdeaProjects/  NTUSER.DAT{3c118866-b738-11ef-b2d7
Contacts/              Links/        NTUSER.DAT{3c118866-b738-11ef-b2d7
Cookies@              'Local Settings'@  NetHood@
DOSBox-0.74/          Music/         OneDrive/

dhrum@DhrumilLathiya MINGW64 ~
$ cd Desktop
bash: cd: Desktop: No such file or directory

```

Figure 6: Inspecting available folders with the ls command.

Description: Lists home-directory contents with ls; cd Desktop fails because Desktop is not in the current folder.

```

dhrum@DhruMilLathiya MINGW64 ~
$ cd OneDrive

dhrum@DhruMilLathiya MINGW64 ~/OneDrive
$ cd Desktop

dhrum@DhruMilLathiya MINGW64 ~/OneDrive/Desktop
$ cd Car Dealership Inventory System_1
bash: cd: too many arguments

dhrum@DhruMilLathiya MINGW64 ~/OneDrive/Desktop
$ cd CarDealershipInventorySystem_1
bash: cd: CarDealershipInventorySystem_1: No such file or directory

dhrum@DhruMilLathiya MINGW64 ~/OneDrive/Desktop
$ cd "Car Dealership Inventory System_1"
bash: $'\302\226cd': command not found

dhrum@DhruMilLathiya MINGW64 ~/OneDrive/Desktop
$ cd "Car Dealership Inventory System_1"

```

Figure 7: Quoting a folder name containing spaces.

Description: Shows path errors and a corrected cd command with a quoted folder name; successful navigation is not yet shown.

```

dhrum@DhruMilLathiya MINGW64 ~/OneDrive/Desktop/Car Dealership Inventory System_1 (main)
$ git log
commit 6dcefe78c90a4435c75794eceaf187cbb1e6f6e6 (HEAD -> main, origin/main, origin/HEAD)
Author: DhruMilLathiya <dhruMilLathiya19@gmail.com>
Date: Sat Jul 11 17:52:01 2026 +0530

docs: Add detailed README file containing API documentation

Written comprehensive setup walkthrough, local configuration files,
test commands, and developer co-authored AI attributions logs based
on the project specifications and constraints.

Co-authored-by: Antigravity <antigravity@users.noreply.github.com>

commit 3d84e22c961f2ab2790aea44cd47dab9f6461a1b
Author: DhruMilLathiya <dhruMilLathiya19@gmail.com>
Date: Sat Jul 11 17:43:44 2026 +0530

docs: Add detailed README file containing API documentation

Written comprehensive setup walkthrough, local configuration files,
test commands, and developer co-authored AI attributions logs.
ChatGPT generated file skeleton structure patterns.

Co-authored-by: ChatGPT <chatgpt@users.noreply.github.com>

commit 87be02391063d1db7fa3c8424c5b44944e8b6cc9
Author: DhruMilLathiya <dhruMilLathiya19@gmail.com>
Date: Sat Jul 11 17:43:27 2026 +0530

```

Figure 8: Viewing the existing project's commit history.

Description: Uses git log inside the project repository to display existing commits, authors, dates, messages, and branch references.

```

commit 2eb9456fd736dc458e5bc6c2b80bdda2e0b633a4
Author: DhrumilLathiya <dhrumillathiya19@gmail.com>
Date: Sat Jul 11 17:43:14 2026 +0530

    feat: Design visual dashboard dashboard grids interface

    Wired search and price-range sliders triggers filters alongside
    showroom list layouts grid display. Antigravity designed dynamic filtering methods.

    Co-authored-by: Antigravity <antigravity@users.noreply.github.com>

commit e7d2b76b96cf738a109a732e08fc3694bba19b20
Author: DhrumilLathiya <dhrumillathiya19@gmail.com>
Date: Sat Jul 11 17:43:00 2026 +0530

    feat: Design registration interface page

    Created user registration forms with security validations warnings.
    Antigravity generated input scopes layout template.

    Co-authored-by: Antigravity <antigravity@users.noreply.github.com>

commit 53074b55326101c1a12dfe6a10d723189b0f5016
Author: DhrumilLathiya <dhrumillathiya19@gmail.com>
Date: Sat Jul 11 17:42:47 2026 +0530

    feat: Design responsive Login page authentication page

```

Figure 9: Inspecting earlier feature-development commits.

Description: Displays earlier commits describing dashboard, registration, and login-interface development, with author and co-author details.

```

dhrum@dhrumilLathiya MINGW64 ~/OneDrive/Desktop/Car Dealership Inventory System_1 (main)
$ q
bash: q: command not found

dhrum@dhrumilLathiya MINGW64 ~/OneDrive/Desktop/Car Dealership Inventory System_1 (main)
$ pwd
/c/Users/dhrum/OneDrive/Desktop/Car Dealership Inventory System_1

dhrum@dhrumilLathiya MINGW64 ~/OneDrive/Desktop/Car Dealership Inventory System_1 (main)
$ cd..
bash: cd.: command not found

dhrum@dhrumilLathiya MINGW64 ~/OneDrive/Desktop/Car Dealership Inventory System_1 (main)
$ cd ..

dhrum@dhrumilLathiya MINGW64 ~/OneDrive/Desktop
$ mkdir food-guests

dhrum@dhrumilLathiya MINGW64 ~/OneDrive/Desktop
$ cd food-guests

dhrum@dhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests
$ mkdir main test

```

Figure 10: Navigating out of the existing project and creating a new folder.

Description: Corrects navigation errors, creates and enters food-guests, then enters mkdir main test to create subfolders.

```

dhru@DhruMiliLathiya MINGW64 ~/OneDrive/Desktop/food-guests
$ ls -R
.:
main/ test/

./main:

./test:

dhru@DhruMiliLathiya MINGW64 ~/OneDrive/Desktop/food-guests
$ pwd
/c/Users/dhru/OneDrive/Desktop/food-guests

dhru@DhruMiliLathiya MINGW64 ~/OneDrive/Desktop/food-guests
$ git init
Initialized empty Git repository in C:/Users/dhru/OneDrive/Desktop/food-guests/.git/

dhru@DhruMiliLathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ git status
On branch main

No commits yet

nothing to commit (create/copy files and use "git add" to track)

```

Figure 11: Initializing the food-guests repository and checking its status.

Description: Initializes food-guests with git init; git status confirms branch main with no commits or tracked files.

```

dhru@DhruMiliLathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ git status
bash: git: command not found

dhru@DhruMiliLathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  README.md

nothing added to commit but untracked files present (use "git add" to track)

dhru@DhruMiliLathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ git add README.md

dhru@DhruMiliLathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ git status
On branch main

No commits yet

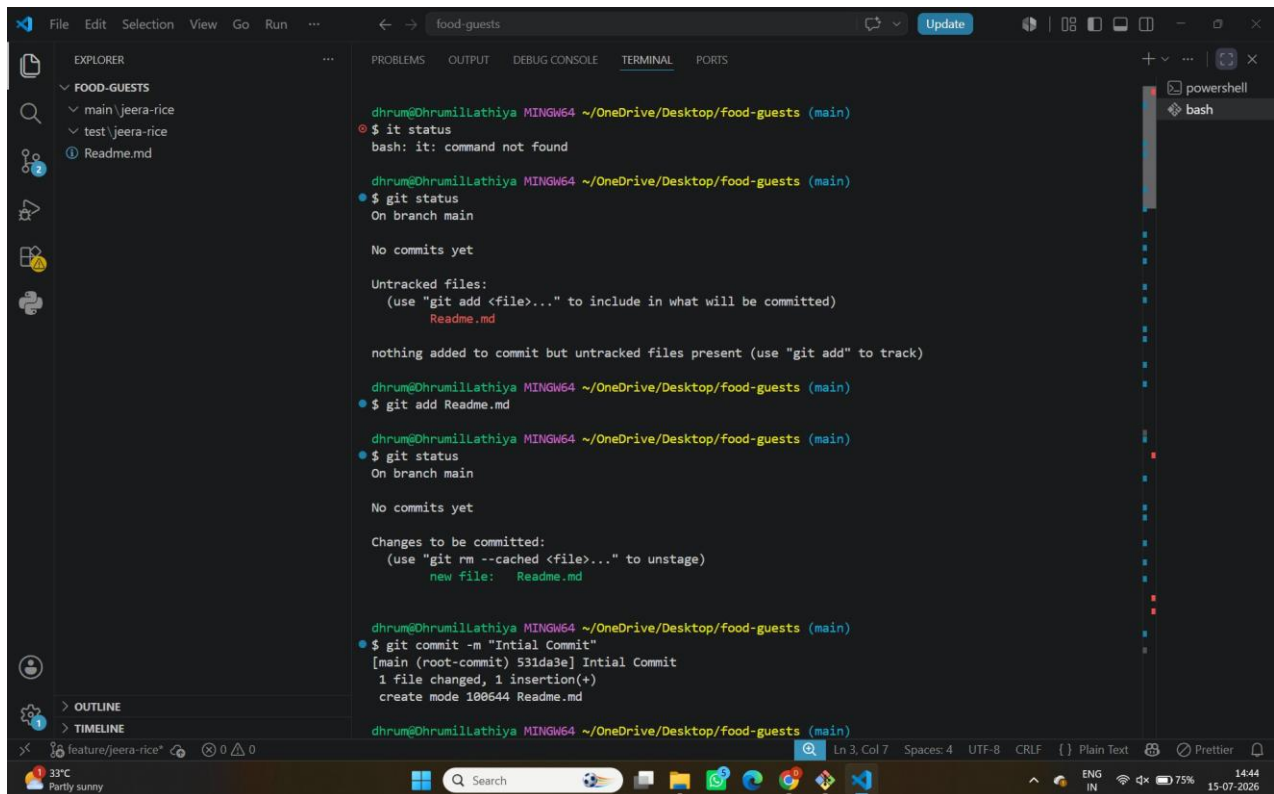
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
  new file:   README.md

dhru@DhruMiliLathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ git commit -m "Initial Commit"
[main (root-commit) 531da3e] Initial Commit
1 file changed, 1 insertion(+)
create mode 100644 README.md

```

Figure 12: Staging the README and creating the initial commit.

Description: Stages README.md with git add and creates the initial commit 531da3e, recording one file and one insertion.



The screenshot shows the Visual Studio Code interface with a terminal window open. The terminal displays the following commands and output:

```
dhru@Dhruillathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ it status
bash: it: command not found

dhru@Dhruillathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    README.md

nothing added to commit but untracked files present (use "git add" to track)

dhru@Dhruillathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ git add README.md

dhru@Dhruillathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   README.md

dhru@Dhruillathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ git commit -m "Intial Commit"
[main (root-commit) 531da3e] Intial Commit
1 file changed, 1 insertion(+)
create mode 100644 README.md

dhru@Dhruillathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
```

The VS Code interface includes the Explorer sidebar on the left showing the project structure (FOOD-GUESTS, main\jeera-rice, test\jeera-rice, README.md), the Output/Debug Console/TERMINAL tabs at the top, and the Windows taskbar at the bottom.

Figure 13: Wider VS Code view of the successful commit workflow.

Description: Shows the VS Code project tree and the same successful README staging and initial commit workflow; no remote push is shown.

```
dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ it status
bash: it: command not found

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    README.md

nothing added to commit but untracked files present (use "git add" to track)

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ git add README.md
```

Figure 14: Close-up of the untracked file and staging command.

Description: Shows Readme.md as untracked and the command `git add Readme.md`; later staging and commit results are outside this crop.

```

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
● $ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   Readme.md

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
● $ git commit -m "Intial Commit"
[main (root-commit) 531da3e] Intial Commit
 1 file changed, 1 insertion(+)
 create mode 100644 Readme.md

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
● $ git log
commit 531da3e94e1a4350b6943412bcaa4e05a0865232 (HEAD -> main)
Author: Dhrumil Lathiya <dhrumillathiya19@gmail.com>
Date:   Wed Jul 15 14:26:49 2026 +0530

    Intial Commit

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
● $ git log --oneline
531da3e (HEAD -> main) Intial Commit

```

Figure 15: Verifying the initial commit in detailed and compact logs.

Description: Creates the initial README commit, then uses `git log` and `git log --oneline` to verify its details on main.

```

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ git log --oneline
531da3e (HEAD -> main) Intial Commit

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (main)
$ git checkout -b feature/jeera-rice
Switched to a new branch 'feature/jeera-rice'

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$ git log
commit 531da3e94e1a4350b6943412bcaa4e05a0865232 (HEAD -> feature/jeera-rice, main)
Author: Dhrumil Lathiya <dhrumillathiya19@gmail.com>
Date:   Wed Jul 15 14:26:49 2026 +0530

    Intial Commit

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$ git log --oneline
531da3e (HEAD -> feature/jeera-rice, main) Intial Commit

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$ git status
On branch feature/jeera-rice
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    main/

nothing added to commit but untracked files present (use "git add" to track)

```

Figure 16: Creating and switching to a feature branch.

Description: Uses git checkout -b feature/jeera-rice to create and switch branches; the log shows the shared initial commit.

```

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$ git add .

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$ git status
On branch feature/jeera-rice
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   main/jeera-rice/recipe.txt

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$ git commit -m "feat(jeera-rice):add steps of preparing Jeera Rice"
[feature/jeera-rice dccd1d1] feat(jeera-rice):add steps of preparing Jeera Rice
1 file changed, 5 insertions(+)
create mode 100644 main/jeera-rice/recipe.txt

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$ git log
commit dccd1d1165f934ebe4c5ed56bb0083990077195e (HEAD -> feature/jeera-rice)
Author: Dhrumil Lathiya <dhrumillathiya19@gmail.com>
Date:   Wed Jul 15 14:34:12 2026 +0530

    feat(jeera-rice):add steps of preparing Jeera Rice

commit 531da3e94e1a4350b6943412bcaa4e05a0865232 (main)
Author: Dhrumil Lathiya <dhrumillathiya19@gmail.com>
Date:   Wed Jul 15 14:26:49 2026 +0530

    Intial Commit

```

Figure 17: Committing the recipe on the feature branch.

Description: Stages the recipe with `git add .` and commits five inserted lines; the log shows the feature branch one commit ahead of main.

```

dhrum@Dhrumillathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$

dhrum@Dhrumillathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$ git log --oneline
dccc1d1 (HEAD -> feature/jeera-rice) feat(jeera-rice):add steps of preparing Jeera
531da3e (main) Intial Commit

dhrum@Dhrumillathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$ git Status
git: 'Status' is not a git command. See 'git --help'.

The most similar command is
    status

dhrum@Dhrumillathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$ git status
On branch feature/jeera-rice
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    test/

nothing added to commit but untracked files present (use "git add" to track)

dhrum@Dhrumillathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$ git add .

```

Figure 18: Correcting command capitalization and staging test files.

Description: Corrects git Status to git status, which lists untracked files under test/, then enters git add . to stage them.

```

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
• $ git status
On branch feature/jeera-rice
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   test/jeera-rice/reciepeTest.txt

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
• $ git commit -m "Test(Jeera-rice):add test for jeera rice"
[feature/jeera-rice aea66af] Test(Jeera-rice):add test for jeera rice
1 file changed, 3 insertions(+)
create mode 100644 test/jeera-rice/reciepeTest.txt

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
• $ git status
On branch feature/jeera-rice
nothing to commit, working tree clean

dhrum@DhrumilLathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
• $ git log --oneline
aea66af (HEAD -> feature/jeera-rice) Test(Jeera-rice):add test for jeera rice
dccc1d1 feat(jeera-rice):add steps of preparing Jeera Rice
531da3e (main) Intial Commit

```

Figure 19: Committing the test file and confirming a clean working tree.

Description: Commits the staged test file with three inserted lines; git status confirms a clean working tree, and the log lists three commits.

```
dhru@Dhruillathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$ git checkout main
Deletion of directory 'main/jeera-rice' failed. Should I try again? (y/n)

dhru@Dhruillathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$ git checkout main
Deletion of directory 'main/jeera-rice' failed. Should I try again? (y/n) y
Deletion of directory 'main/jeera-rice' failed. Should I try again? (y/n) y
Deletion of directory 'main/jeera-rice' failed. Should I try again? (y/n) y
Deletion of directory 'main/jeera-rice' failed. Should I try again? (y/n) y

dhru@Dhruillathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$ git branch
* feature/jeera-rice
  main

dhru@Dhruillathiya MINGW64 ~/OneDrive/Desktop/food-guests (feature/jeera-rice)
$
```

Figure 20: Unsuccessful checkout of the main branch.

Description: Attempts to switch to main fail with directory-deletion errors; git branch confirms feature/jeera-rice remains active.